

Source Tree Manifest

Companion to `04-source-code/README.md` (Deliverable #4)

This file annotates every significant tracked path in the repository at the submission tag `accreditation-2026-06-03`, one line per path, so DOTr reviewers can navigate the tree without guessing what each file is for. The full list of tracked files can always be regenerated with `git ls-files`.

Paths excluded from this manifest: build outputs (`build/`, `dist/`, `target/`, `.gradle/`), package caches (`node_modules/`, `.venv/`), IDE state (`.idea/`, `.vscode/`), the local SQLite database, the local photos directory, and Claude session metadata. These are all in `.gitignore` and are absent from the read-only clone.

Top level

Path	Description
<code>README.md</code>	Repo overview, architecture summary, dev quickstart.
<code>LICENSE.md</code>	Proprietary licence governing this accreditation access grant.
<code>Makefile</code>	Cross-component dev shortcuts (build / test / run for desktop + cloud).
<code>docker-compose.yml</code>	Local Postgres + Redis + MinIO services for cloud dev.
<code>.gitignore</code>	Excluded paths (caches, builds, secrets, local DB, local photos).

`desktop/` — center-side desktop application

`desktop/` shell

Path	Description
<code>desktop/package.json</code>	Electron + renderer NPM package; entrypoints, build scripts.
<code>desktop/pyproject.toml</code>	Python sidecar package; dependencies, entry script <code>petc-sidecar</code> .
<code>desktop/requirements.txt</code>	Locked Python deps (mirrored from pyproject for installer).

`desktop/electron/` — Electron main process

Path	Description
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electron/main.ts	Main process — spawns sidecar, opens renderer window.
electron/preload.ts	Preload script — exposes window.petcBridge API to the renderer.
electron/bridge.d.ts	Type declarations for the renderer ↔ main IPC bridge.
electron/tsconfig.json	TypeScript config for the main process.

desktop/renderer/ — React UI

Path	Description
renderer/package.json	Renderer NPM package (React, Vite, Tailwind, Vitest).
renderer/vite.config.ts	Vite build config.
renderer/tailwind.config.js	Tailwind theme.
renderer/src/main.tsx	React entry point.
renderer/src/App.tsx	Top-level component + router.
renderer/src/api/sidecarClient.ts	Typed wrapper around the local sidecar HTTP API.
renderer/src/components/AppShell.tsx	Layout (nav bar, status bar).
renderer/src/components/CameraStream.tsx	Live camera preview + capture.
renderer/src/pages/auth/LoginPage.tsx	Operator login.
renderer/src/pages/test/RunTestPage.tsx	Run-test flow (start, capture readings, complete).
renderer/src/pages/upload/LtmsUploadPage.tsx	Six-step upload wizard — the primary submission UI.
renderer/src/pages/history/HistoryPage.tsx	Test history table with CEC print + WAITING_FOR_LTMS polling.
renderer/src/pages/analytics/AnalyticsPage.tsx	Per-center analytics.
renderer/src/pages/settings/SettingsPage.tsx	Analyser / camera / printer settings.
renderer/src/store/authStore.ts	Zustand store for the logged-in operator.
renderer/src/types/index.ts	Shared TS types mirroring the sidecar API.
renderer/src/utils/emissionLimits.ts	Verdict computation (PASS / FAIL with reasons).

desktop/sidecar/ — Python FastAPI sidecar

Path	Description
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sidecar/petc/service.py	Startup — wires analyser, camera, printer, gov client, sync threads.
sidecar/petc/cloud_client.py	HTTP client for the cloud /api/photos/presign , /api/submissions , registry endpoints.
sidecar/petc/analyser/base.py	Analyzer protocol + FuelType + reading dataclasses.
sidecar/petc/analyser/builder.py	Constructs the configured analyser from settings.
sidecar/petc/analyser/serial_base.py	Common base for USB-Serial analysers.
sidecar/petc/analyser/fofen_gas.py	Fofen gas analyser (binary protocol).
sidecar/petc/analyser/fofen_ascii.py	Fofen gas analyser (ASCII receipt mode).
sidecar/petc/analyser/fofen_framing.py	Fofen binary framing primitives.
sidecar/petc/analyser/fty_opacimeter.py	FTY-100 diesel opacimeter (ASCII receipt parser).
sidecar/petc/analyser/ascii_gas.py	Generic ASCII gas analyser.
sidecar/petc/analyser/binary_diesel.py	Generic binary diesel analyser.
sidecar/petc/analyser/mock.py	Deterministic mock analyser used in tests and dev.
sidecar/petc/camera/capture.py	Camera capture interface + mock implementation.
sidecar/petc/camera/opencv_camera.py	OpenCV-backed USB webcam capture.
sidecar/petc/camera/builder.py	Selects camera implementation per config.
sidecar/petc/printer/base.py	Printer protocol + ReceiptData .
sidecar/petc/printer/mock.py	Mock printer used in tests and dev.
sidecar/petc/gov/base.py	Local gov adapter protocol (GovRegistryClient) + dataclasses.
sidecar/petc/gov/mock_client.py	Local mock gov client (offline / dev path).
sidecar/petc/gov/stradcom_client.py	Local Stradcom adapter stub.
sidecar/petc/api/server.py	FastAPI app — every desktop HTTP endpoint.
sidecar/petc/cec/pdf.py	CEC PDF renderer (two copies per A4).
sidecar/petc/db/models.py	SQLAlchemy ORM models for the local SQLite DB.
sidecar/petc/db/session.py	SQLAlchemy session factory + DB path resolution.
sidecar/petc/db/migrations/env.py	Alembic env.
sidecar/petc/cloud_sync/pusher.py	Background daemon that drains the cloud mirror outbox.
sidecar/petc/submissions/reconciler.py	Background daemon that resolves WAITING_FOR_LTMS submissions by polling the

	cloud.
sidecar/petc/queue/outbox.py	Generic outbox / retry helper.
sidecar/petc/sync/	Reserved for future sync helpers (empty stub package).

desktop/tests/ – pytest suite

Path	Description
tests/test_api.py	End-to-end tests of the sidecar HTTP API.
tests/test_analyzer.py	Analyser protocol + mock + fixture tests.
tests/test_serial_analyzers.py	Fofen / FTY parser tests using captured byte fixtures.
tests/test_outbox.py	Outbox retry + dead-letter tests.
tests/test_cloud_client.py	cloud_client.py tests using httpx.MockTransport.
tests/fixtures/*.bin, *.txt	Captured analyser frames used by the parser tests.

desktop/installer/

Path	Description
installer/electron-builder.yml	Electron Builder config for the Windows MSI.
installer/petc_sidecar.spec	PyInstaller spec used to freeze the Python sidecar inside the desktop bundle.

cloud/ – Spring Boot cloud service

cloud/ shell

Path	Description
cloud/build.gradle.kts	Gradle build for the active cloud service.
cloud/settings.gradle.kts	Gradle settings.
cloud/Dockerfile	Container image for the cloud service.
cloud/gradlew / gradlew.bat / gradle/wrapper/	Gradle wrapper (8.10).

cloud/src/main/java/com/petc/

Path	Description
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PetcApplication.java	Spring Boot main class (the active entry point).
auth/SecurityConfig.java	HTTP security (filter chain, allowed paths, JWT).
auth/AuthController.java	Operator-portal login + refresh endpoints.
auth/AuthService.java	Login + refresh token issuance.
auth/JwtService.java	JWT signing / parsing.
auth/JwtAuthFilter.java	Reads the JWT, populates the security context.
auth/User.java / UserRepository.java	Operator persistence.
auth/RefreshToken.java / RefreshTokenRepository.java	Refresh-token persistence (revocable).
auth/Role.java / AuthException.java	Role enum + auth-specific errors.
tenant/Tenant.java	Tenant entity.
tenant/TenantAwarePrincipal.java	Carries the tenant claim through the security context.
tenant/TenantContextFilter.java	SET LOCAL app.tenant_id per request — drives Postgres RLS.
config/S3Config.java	S3Client + S3Presigner beans.
common/GlobalExceptionHandler.java	Uniform error responses.
gov/GovRegistryClient.java	Interface for vehicle / driver lookup + submit-emission-result.
gov/MockGovRegistryClient.java	Active when petc.gov.mock=true (the default).
gov/StradcomGovRegistryClient.java	Stub for the real Stradcom integration; active when petc.gov.mock=false.
gov/VehicleInfo.java , DriverInfo.java , EmissionPayload.java , SubmissionResult.java	Gov-side DTOs (records).
ingest/CenterKeyValidator.java	X-Center-Key validation + tenant resolution.
registry/RegistryController.java	/api/registry/vehicle/{plate} and /api/registry/driver/{lic}.
photos/PhotosController.java	/api/photos/presign — short-lived S3 PUT URL.
submissions/SubmissionsController.java	POST /api/submissions , GET /api/submissions/{id}.
submissions/SubmissionService.java	Enqueue, status, state transitions, idempotency.

submissions/SubmissionJobRunner.java	@Scheduled background worker, retries, back-off.
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cloud/src/main/resources/

Path	Description
application.yml	Spring Boot config (datasource, S3, gov adapter, submission retry policy).
db/migration/V1__init.sql	Initial schema (tenants , users , mirror tables, audit log).
db/migration/V2__submissions.sql	center_licenses , submissions table + RLS policy.
db/migration/V3__submission_cec_fields.sql	Adds or_no , dermalog_token , valid_from , valid_until .

cloud/src/test/java/com/petc/

Path	Description
gov/MockGovRegistryClientTest.java	Plate-behaviour tests, cert uniqueness, Optional semantics.
submissions/SubmissionJobRunnerTest.java	Accepted / rejected / gov-throws / empty-batch coverage.

cloud/frontend/ – operator portal (React)

Path	Description
frontend/package.json	React + Vite + Tailwind operator portal.
frontend/src/...	Login, dashboard, tenant management, history browsing.
frontend/nginx.conf	Production reverse proxy config for the portal container.
frontend/Dockerfile	Container image for the portal.

cloud/backend/ – DEPRECATED older spike

Path	Description
cloud/backend/**	Older Spring Boot spike. Not the production cloud. See <code>cloud/backend/DEPRECATED.md</code> .

shared/

Path	Description
shared/contracts/mirror-events.schema.json	Desktop → cloud mirror event envelope.
shared/contracts/presign-request.schema.json	POST /api/photos/presign request schema.
shared/contracts/presign-response.schema.json	POST /api/photos/presign response schema.
shared/contracts/submission-request.schema.json	POST /api/submissions request schema.
shared/contracts/submission-status.schema.json	GET /api/submissions/{id} response schema.
shared/ui/	Shared UI assets (icons, brand).

docs/accreditation/ — this submission package

Path	Description
README.md	Index of all six deliverables and their draft status.
01-client-application-manual.md	Operator-facing client application manual.
02-setup-and-network-layout.md	Centre hardware + center LAN + edge of the WAN.
03-system-documentation.md	Software architecture, security, sub-program list.
04-source-code/README.md	This deliverable — access, build, third-party components.
04-source-code/manifest.md	This file.
05-cec-samples/README.md	CEC sample documentation + field mapping.
05-cec-samples/sample-cec-mock-data.pdf	One rendered sample CEC (mock data).
06-network-architecture.md	AWS / cloud platform architecture.

End of source tree manifest.